COMP4442

SERVICE AND CLOUD COMPUTING

Virtualization
Part-2

Storage Virtualization

Aggregate various heterogeneous storage devices into a pool of shared logical storage, seen as a single logical storage device:

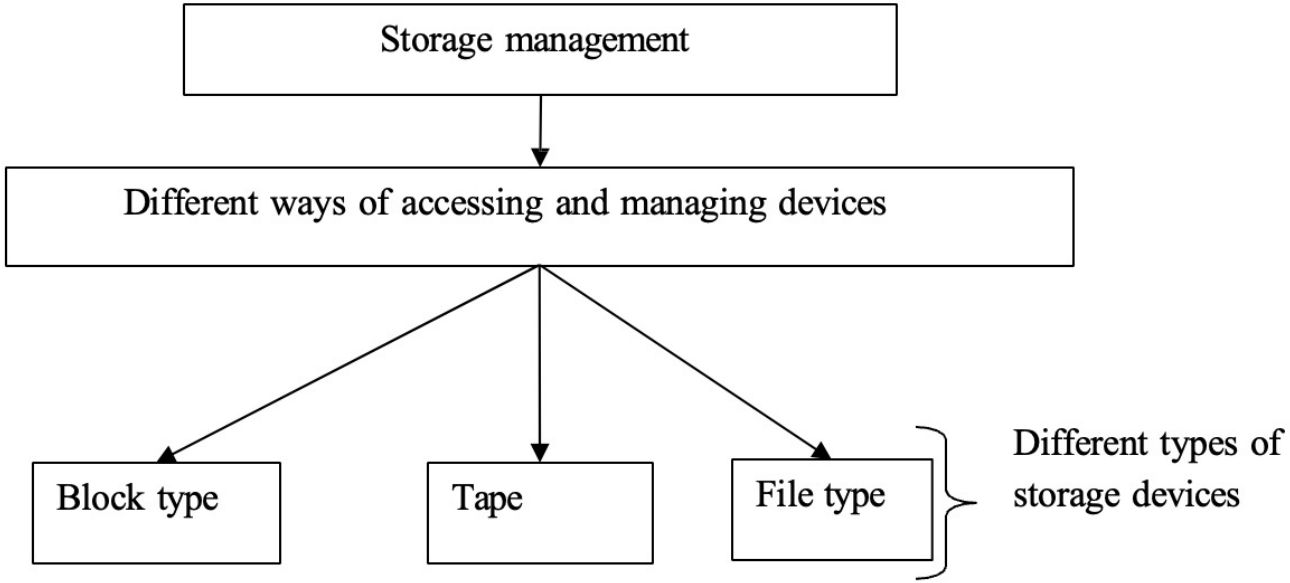
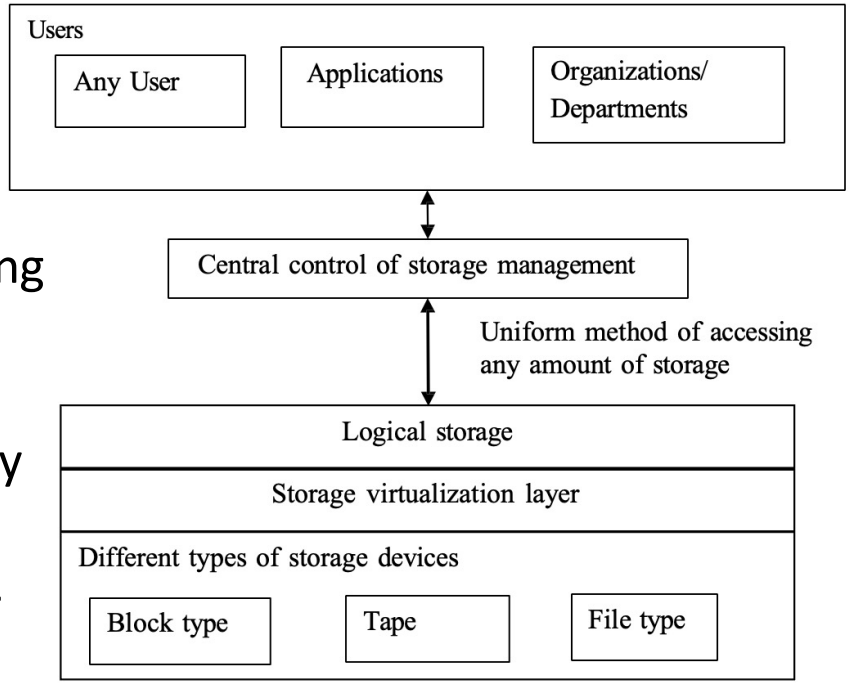


Fig. 3.16 Conventional storage with no virtualization

Hide varying characteristics of different storage devices as well as vendor-specific features

Abstract heterogeneity among different storage devices to provide (automated) management in a unified way

Better storage utilization, higher availability, easy backup, achieving, recovery of the data



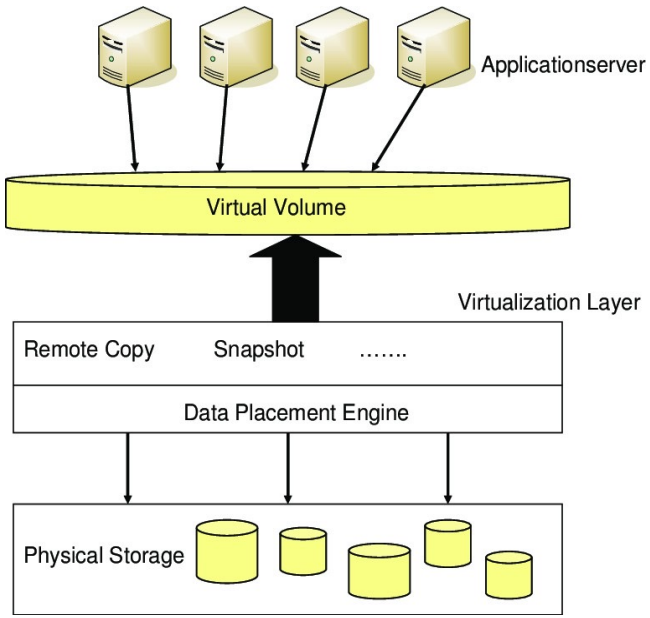


Fig. 3.17 Virtualized storage

Storage Virtualization: Desirable Properties

Manageability

Storage resource should be easily configured and deployed.

Availability

Storage hardware failures should not affect the application.

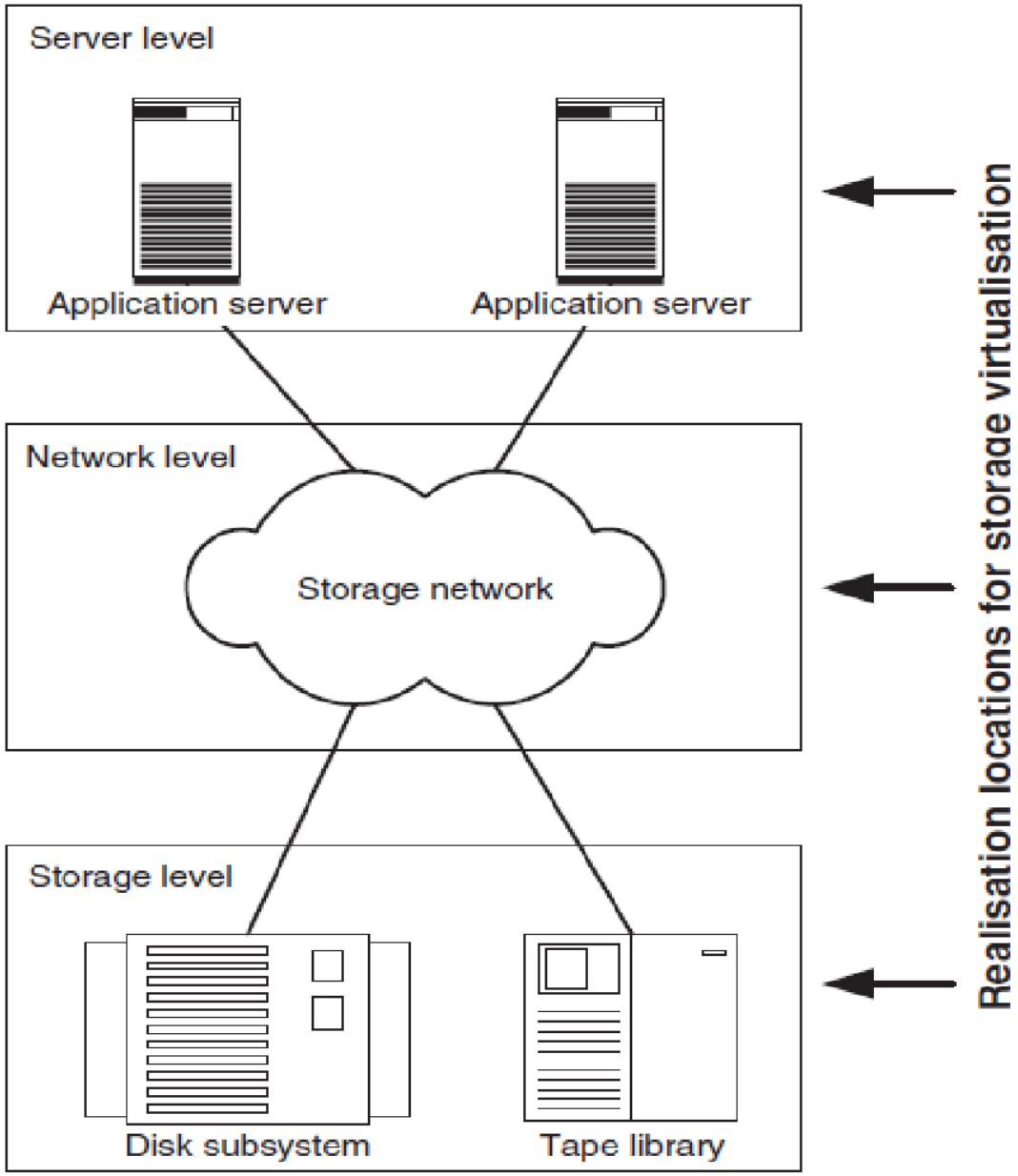
Scalability

Storage resource can easily scale up and down.

Security

Storage resource should be securely isolated.

Storage Virtualization: Realization Levels



Server-level virtualization

All the virtualizations and management is done at the host level, typically, in the form of combination of file system and volume manager

Volume manager separates the storage into logical view and physical implementation by encapsulating the physical hard disk into logical disk groups and logical volumes.

The virtualisation functions can integrate multiple storage systems.

However, storage virtualisation software must take place on every single server

- Can cost system resources

Storage Virtualization: Realization Levels

Network-level virtualization

The most common form used nowadays.

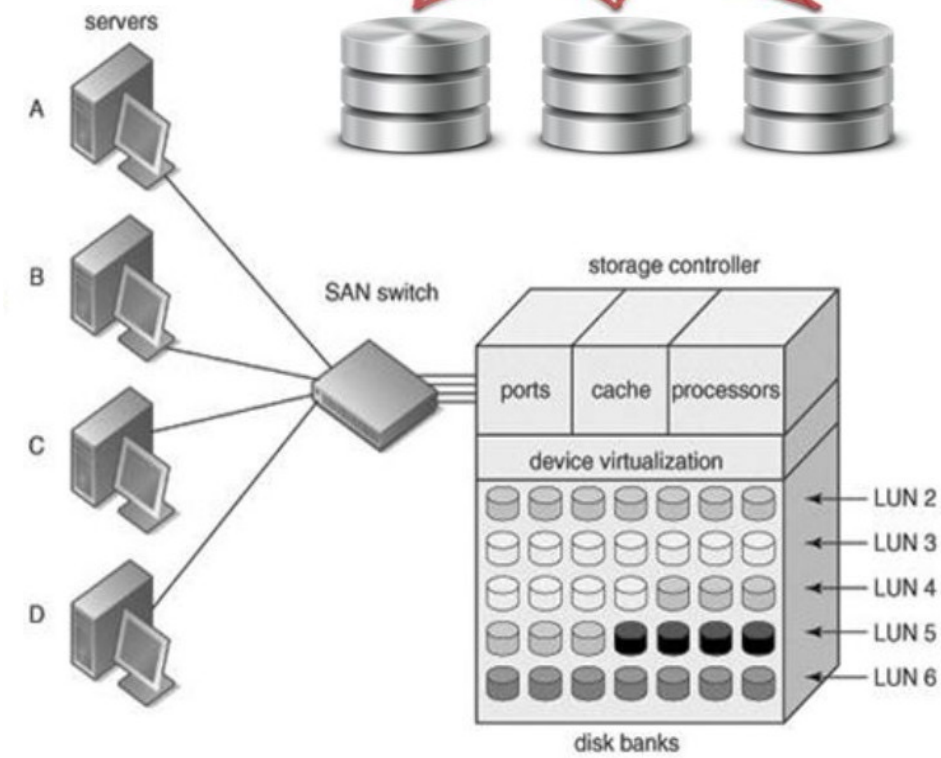
Devices such as a smart switch or purpose-built server connect to all the storage devices in a fibre channel storage network and present the storage as a virtual pool.

Storage array-level virtualization

Storage controller provides different types of physical storage used as storage tiers, which are made up of solid-state drives hard drives.

Cache memory enhances the performance.

Multiple storage systems with coordination are necessary to ensure high availability.



Storage Virtualization: Data-access Levels

File-level Storage Virtualization

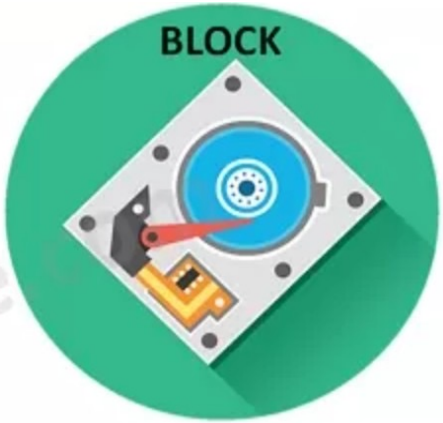
- Applied to network-attached storage (NAS) system
- Breaks the dependency between the data being accessed and the location of the physical memory, using NFS protocols or Server Message Block.
- Enables movement of files between NASs in the background which improves performance.

Block-level Virtual Storage

- Blocks are created for logical storage which are just like a hard drive or any solid state memory device in the server.
- Used and managed by the OS on the servers.
- More widely used than the file-based virtual storage system.

SPECIFIC FOLDER
IN
FIXED LOGICAL ORDER

File path
File name
Date



SPECIFIC LOCATION
ON
DISKS/MEMORY

Tracks
Sectors

BASIS	FILE LEVEL STORAGE	BLOCK LEVEL STORAGE
Deployed for	Network attached storage devices	Storage Area Networks
Storage Protocols	NFS, SMB or CIFS	Fibre Channel, iSCSI and FCoE (Fibre Channel over Ethernet)
Kind of storage system	Centralized file storage	Blocks are created to store files which are utilized as hard drives by the server operating system
File and Folder Access	Can be accessed and managed by this storage system	Not able to manage or control smaller blocks of storage
Implementation	More simple to implement	Complex in usage
Flexibility	Less flexible	More flexible
File sharing	Access permissions can be given to the users for sharing the files	For sharing files, an operating system is to be separately installed and attached to the block
Cost	Less expensive to manage	More expensive to manage
Application	A suitable option when only an empty space is required to dump the files	Larger enterprises use it to store data in the form of blocks and their flexibility
Uses	Used for mass storage of files and VMware	It has more uses as it can store files, databases, VMware.
Server boot	Cannot be done	Server reboots can be done if right kind of devices are used

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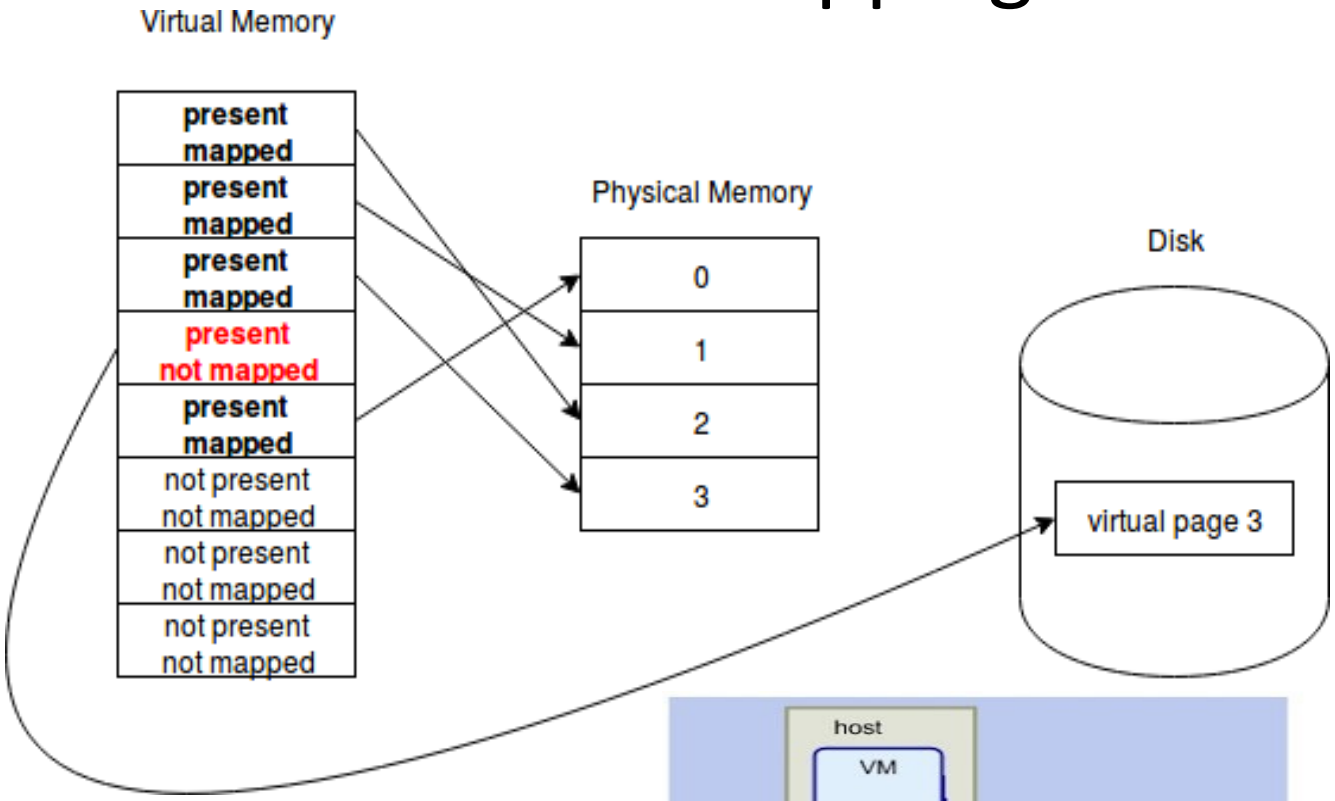
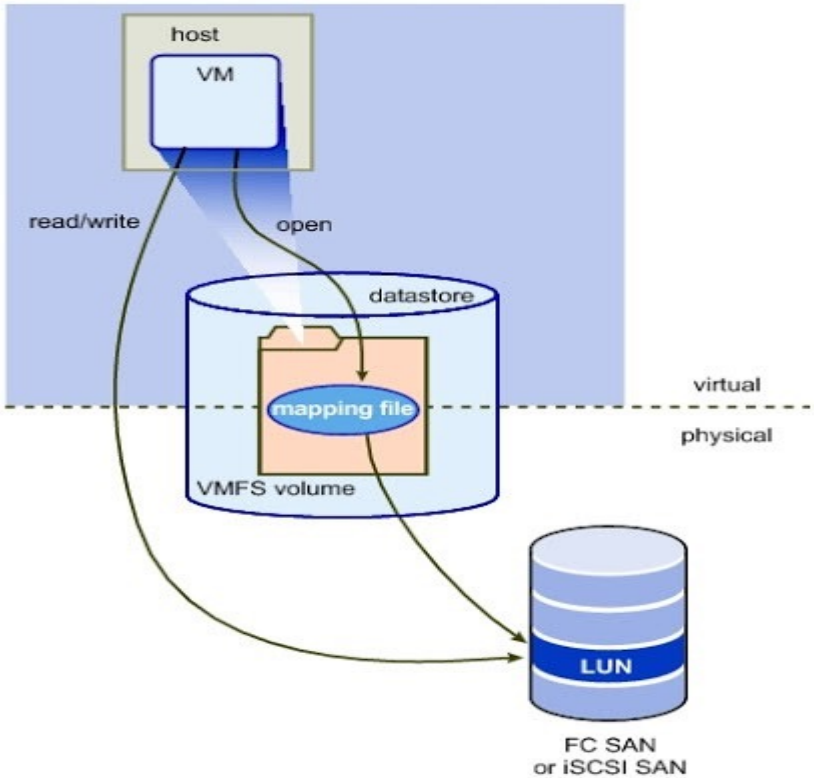
Storage Virtualization: Address Remapping*

Storage virtualization helps to achieve location independence and provides the logical space to the user to store their data.

This requires handling the process of address mapping.

It is possible that the output of virtualizations can cascade as an input for a higher level of virtualizations, leading to the possibility of multiple layers of virtualizations mapping.





Network Virtualization

Create more than one logical / virtual network that work simultaneously, sharing the same single physical network

- Networks can be virtualized with virtual switches and links

Different virtual networks are configured by different virtual machines

- with their own topology, routing, and other application-specific requirements.

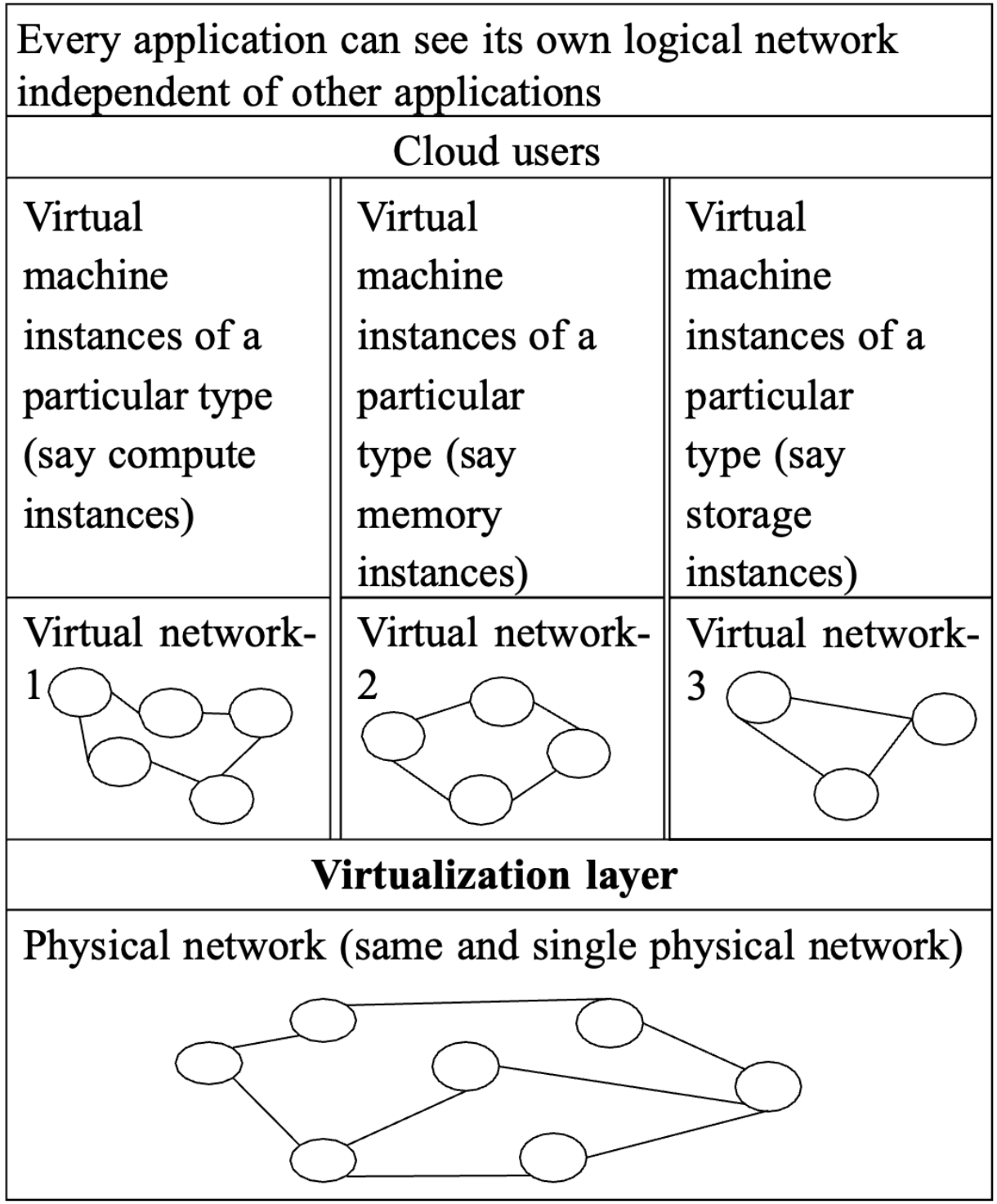
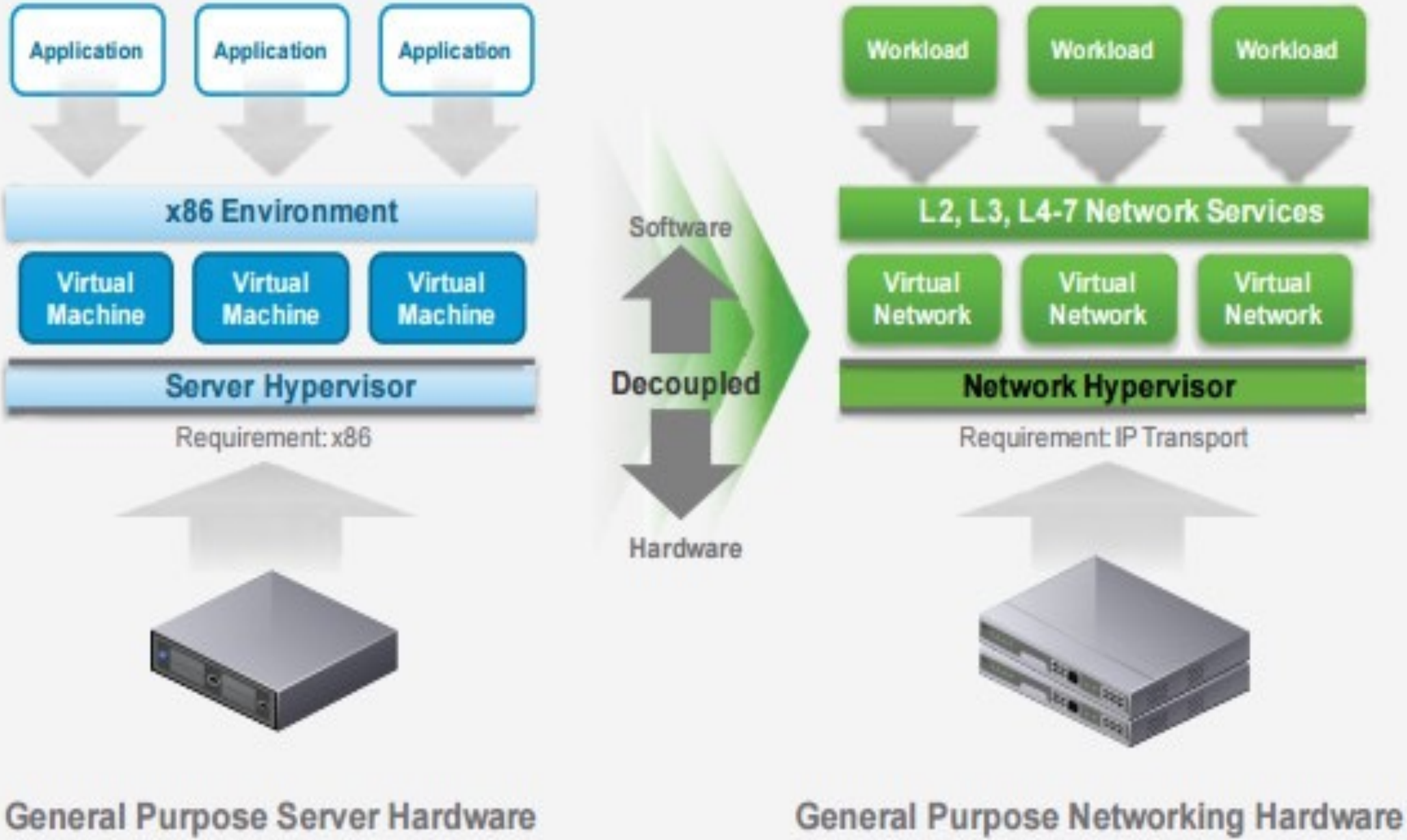


Fig. 3.18 Network virtualization

Network Virtualization



Network Virtualization Is Not New*

VPN: Virtual Private Networks

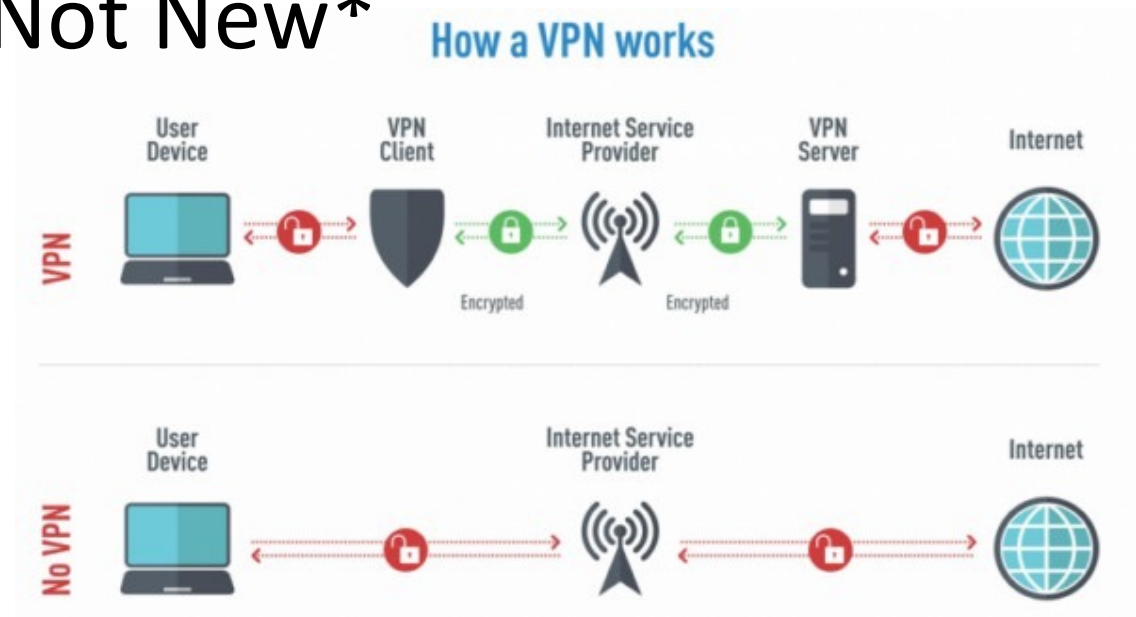
- Virtual network connecting distributed sites
- Not customizable enough

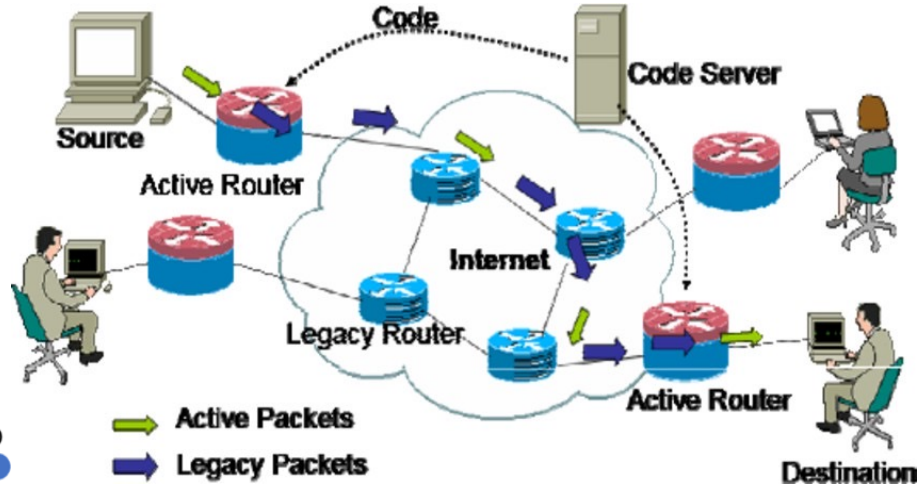
Active and Programmable Networks

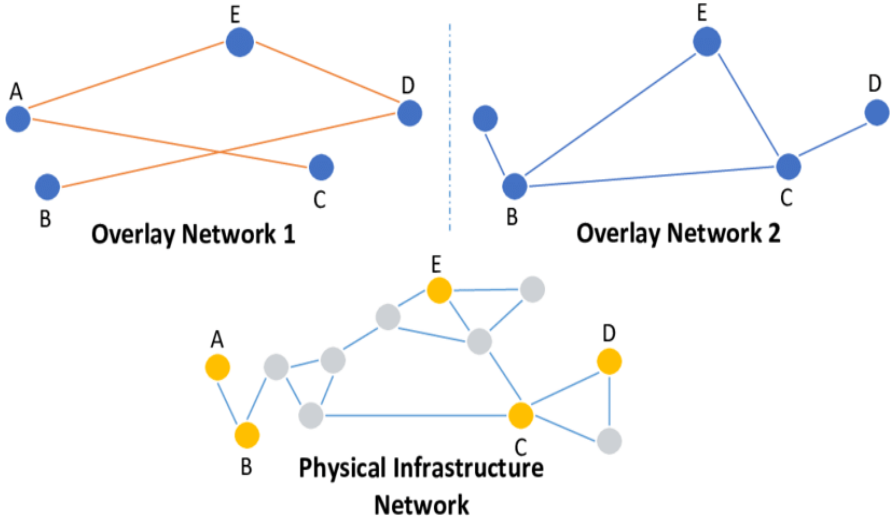
- Customized networks functionalities
- Programmable interfaces and active codes

Overlay Networks

- Application layer virtual networks
- Not flexible enough







Network Virtualization: Desirable Properties

Flexibility

- Service providers can choose arbitrary network topology, routing and forwarding functionalities, customized control and data planes
- No need for coordination with others

Scalability

- Maximize the number of co-existing virtual networks
- Increase resource utilization

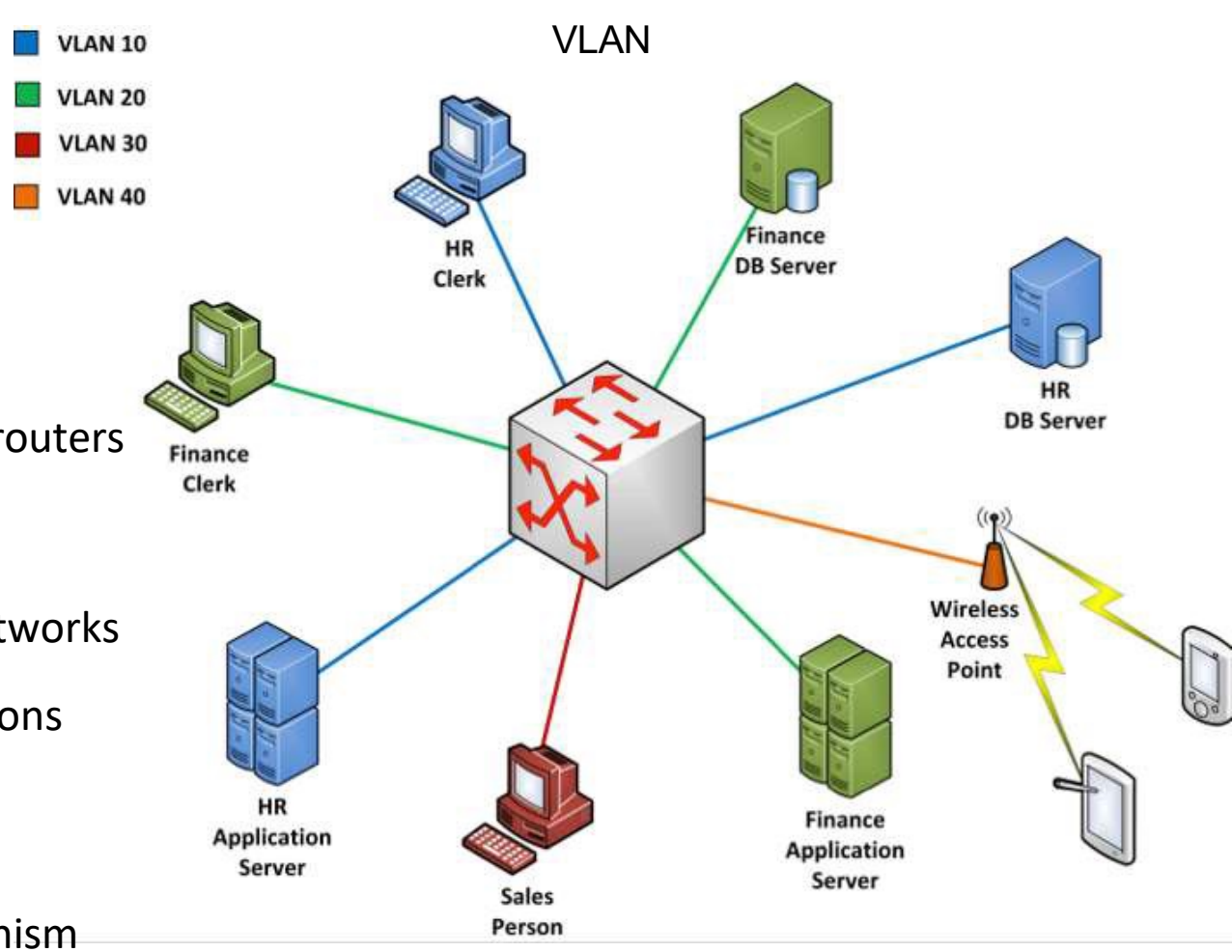
Programmability of network elements like routers

Security, privacy and isolation

- Complete isolation between virtual networks
- Isolate faults, bugs, and misconfigurations

Manageability

- Clear separation of policy from mechanism
- Defined accountability of infrastructure and service providers



Network Virtualization: Objectives*

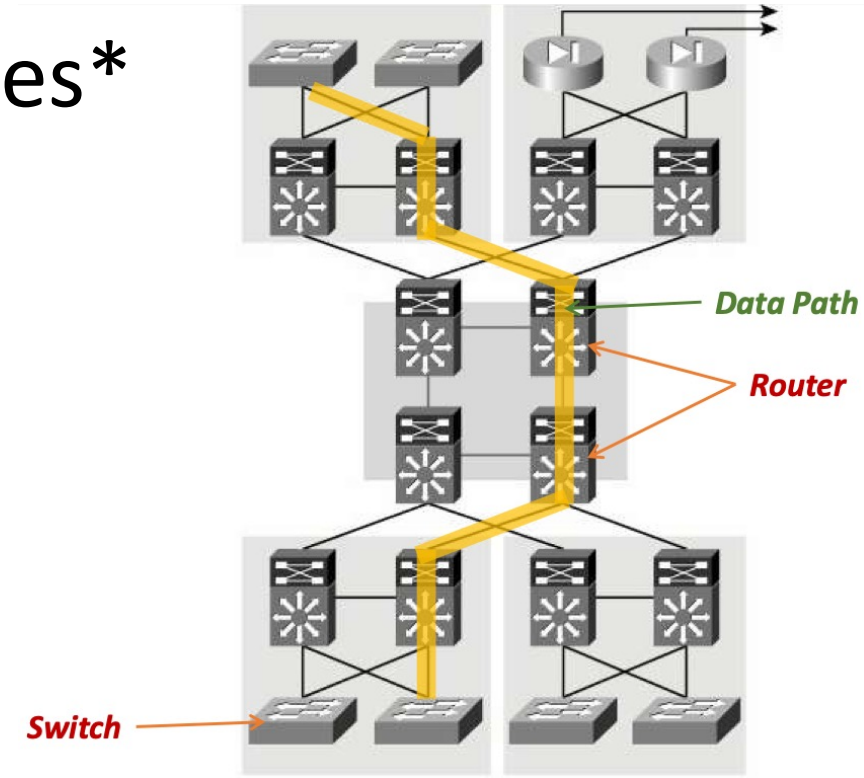
Network device virtualization

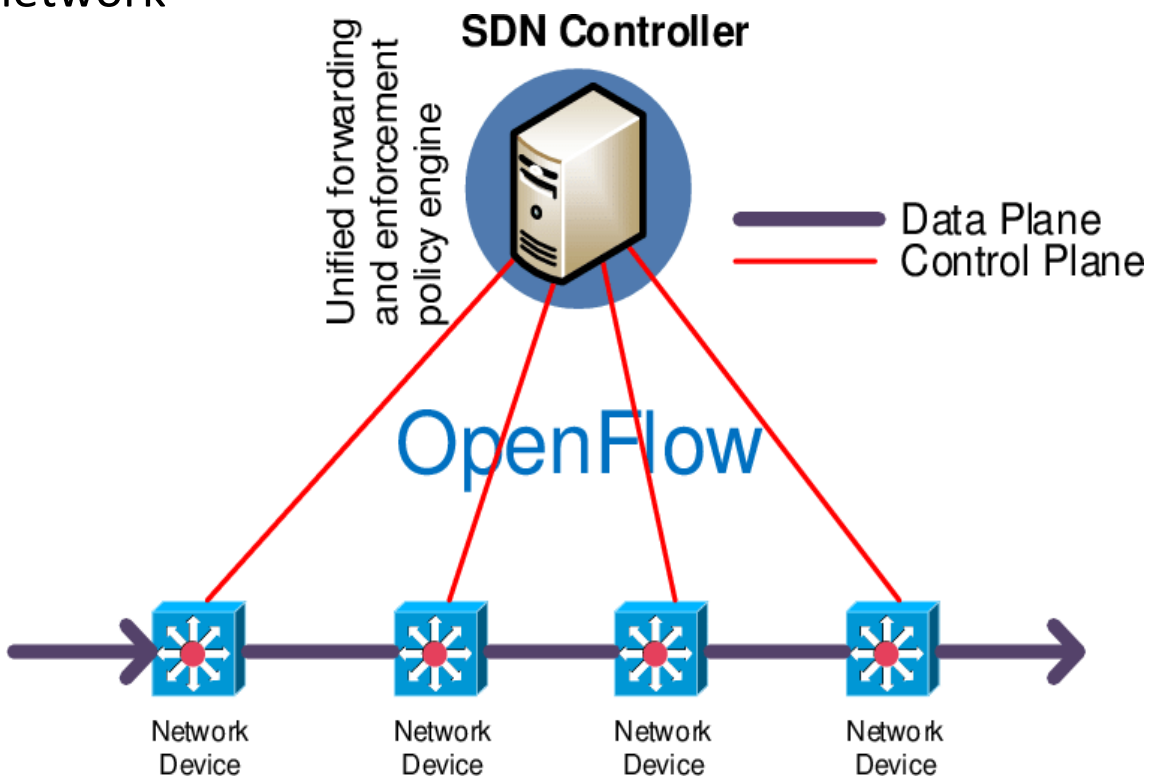
- Virtualize physical devices (nodes) in the network
  - Data Plane virtualization
  - Control Plane virtualization
  - Management Plane virtualization

Network data path virtualization

- Virtualize communication path between network access points
  - Links virtualization

Can be achieved by SDN

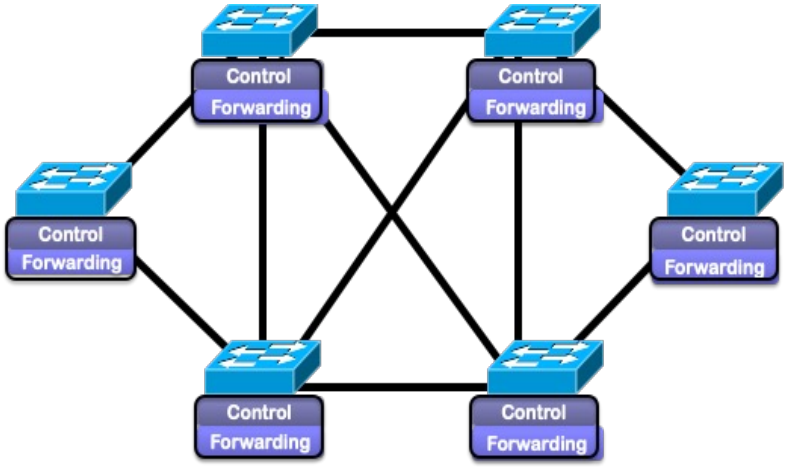




Software Defined Network

Many complex functions baked into the infrastructure

OSPF, BGP, mul,cast, Traffic Engineering, NAT, firewalls, MPLS, …

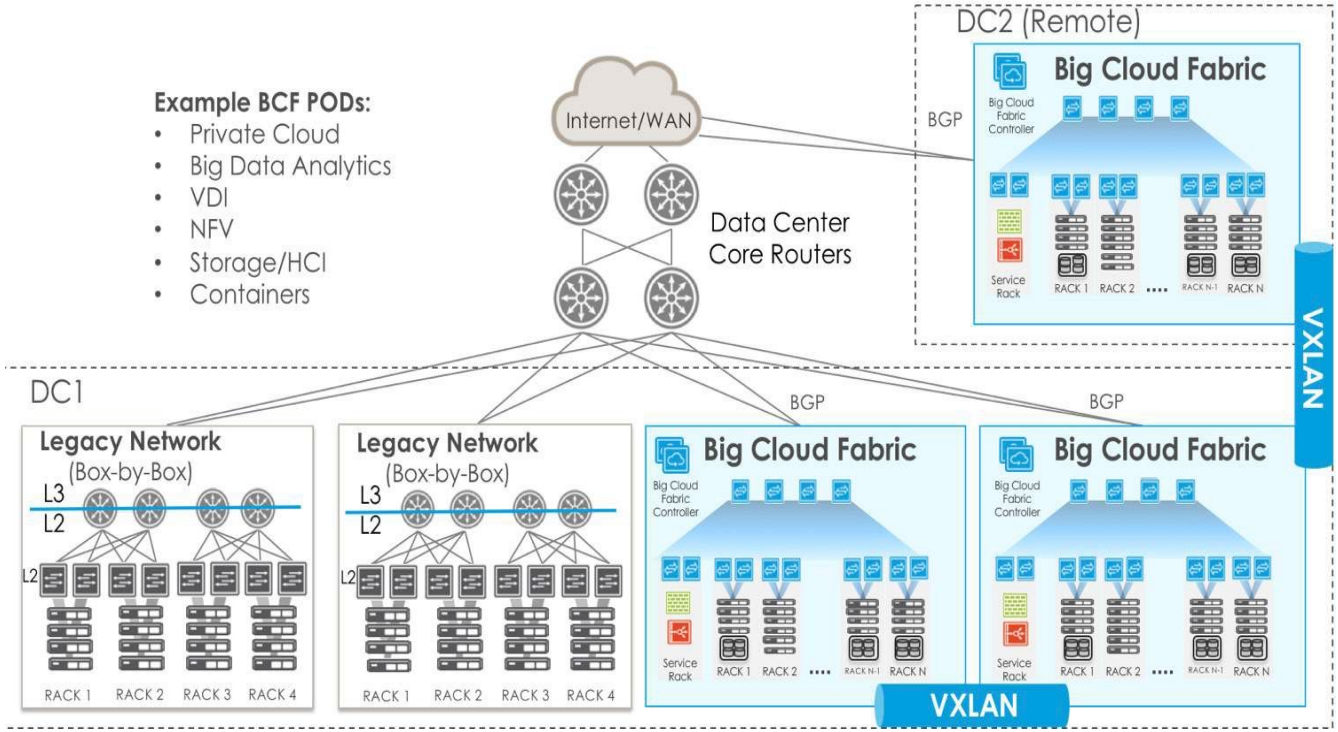


Problems in conventional networks

- buggy software in equipment
- lack of flexibility and agility
- vendor-specific network devices
- high operational expenditure

In cloud computing environment, more problems:

- Networks consist of large number of networking devices and involve complex configurations.
  - Manually configuring devices is tedious and time consuming.
  - Inflexible network configuration, e.g., mobility
- Under-utilization of network resources



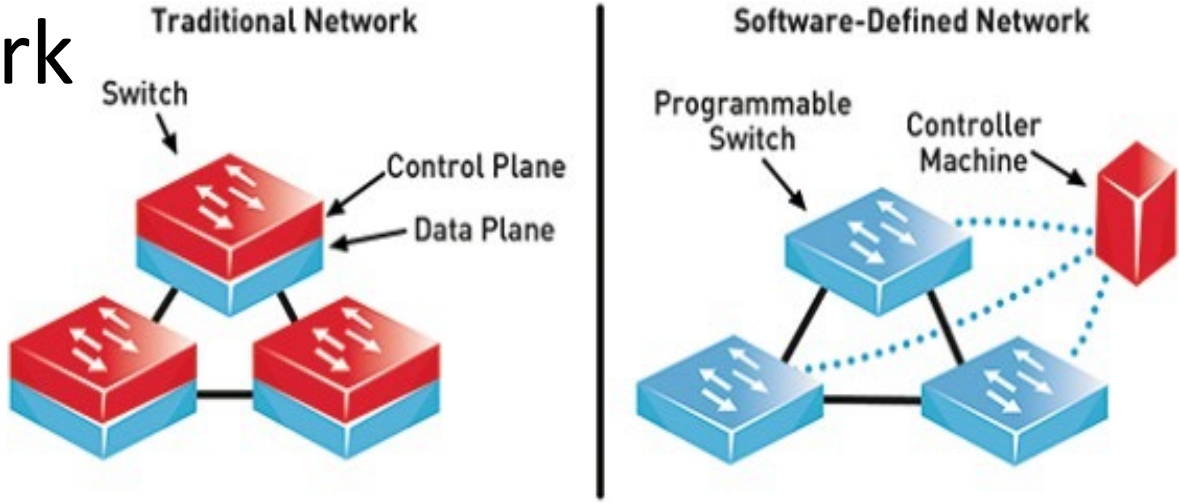
Software Defined Network

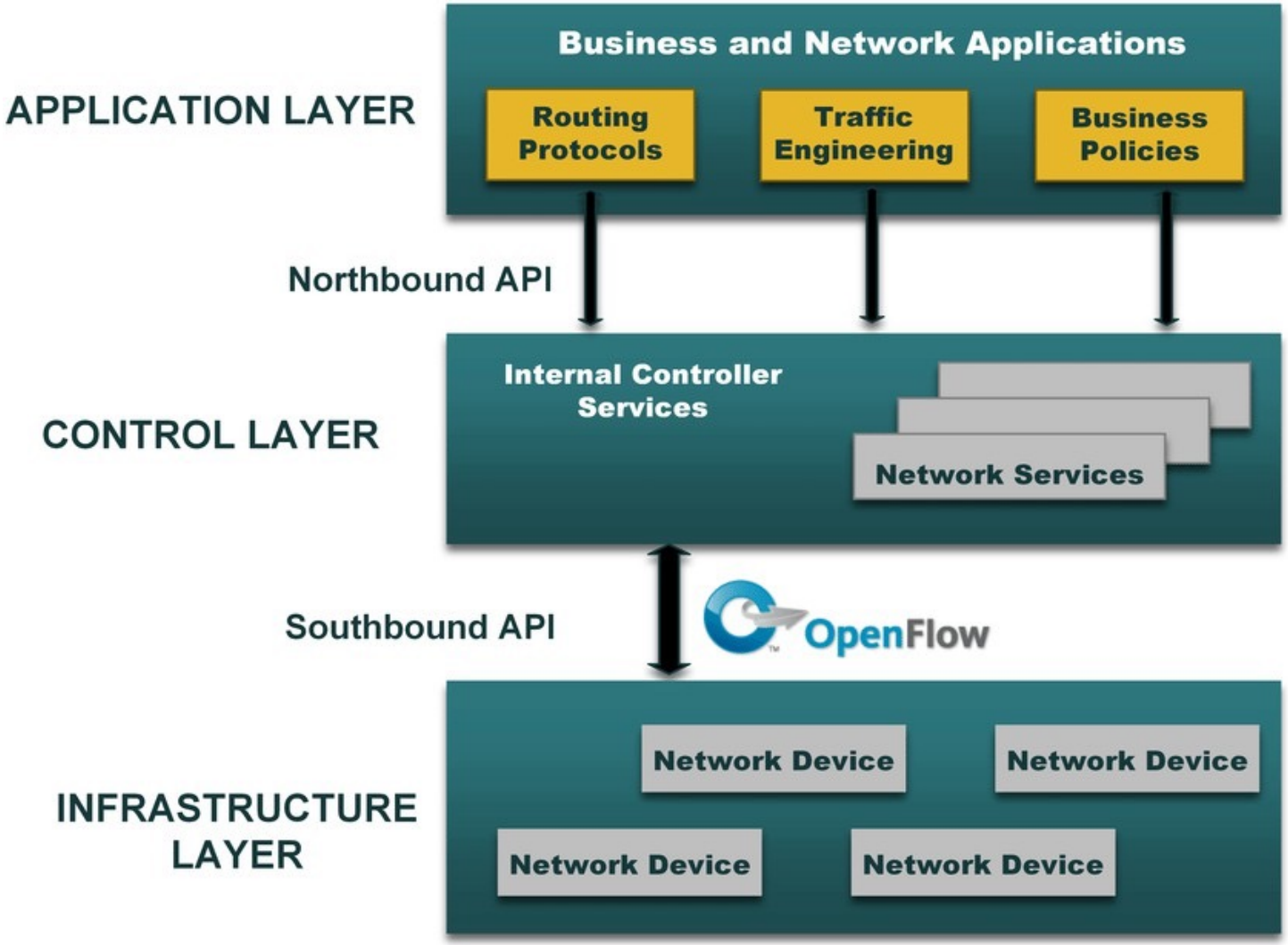
Solutions?

- Separate control plane and data plane
- Centralize network management / control
- Network programmability
- Network virtualization

SDN (Software Defined Networks): a concept and technology used to configure network devices and their functions by means of programmable interfaces.

- Separate the network functionality from the control of the network.
- A logically centralized *controller* provides a global and unified view of the entire network.

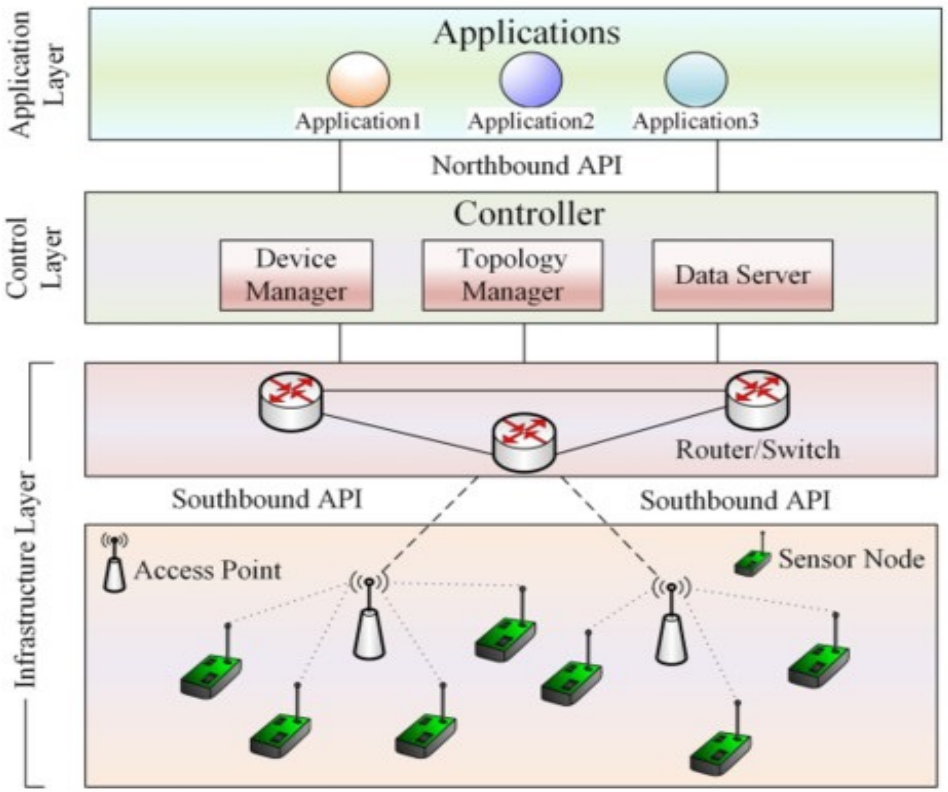


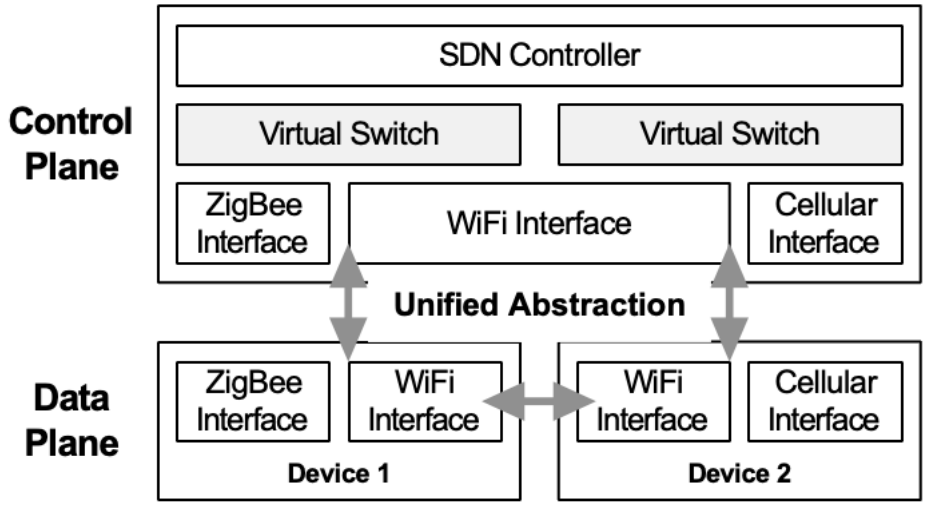


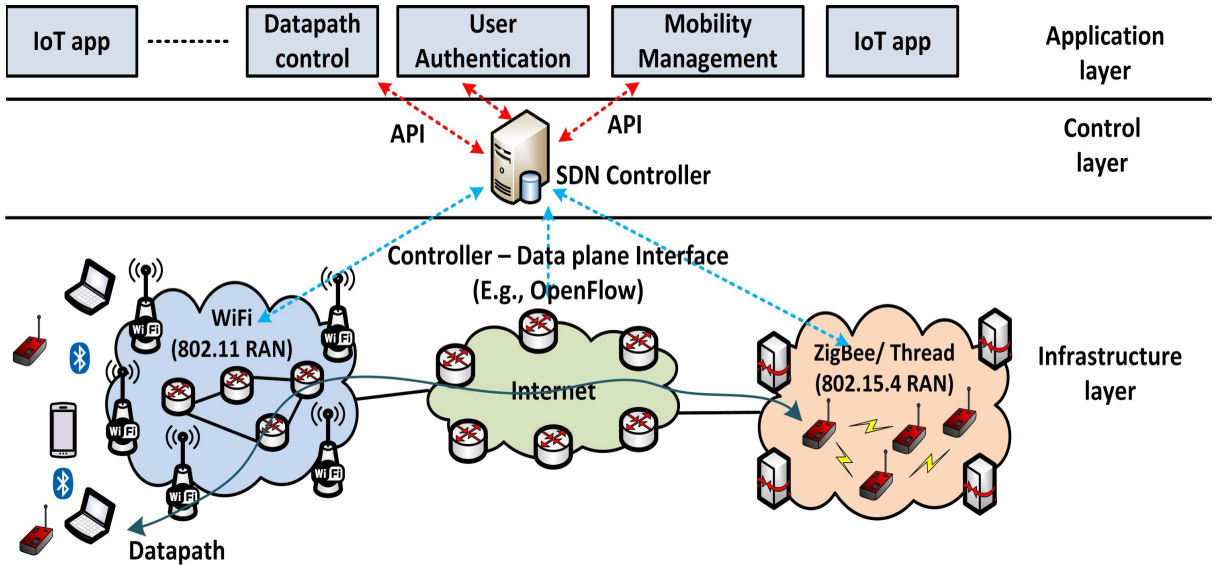
SDN Hides Heterogeneity

SDN can utilize virtual networks to tackle challenges of interoperability, interference and resource provisioning

Providing unified abstraction of different interfaces



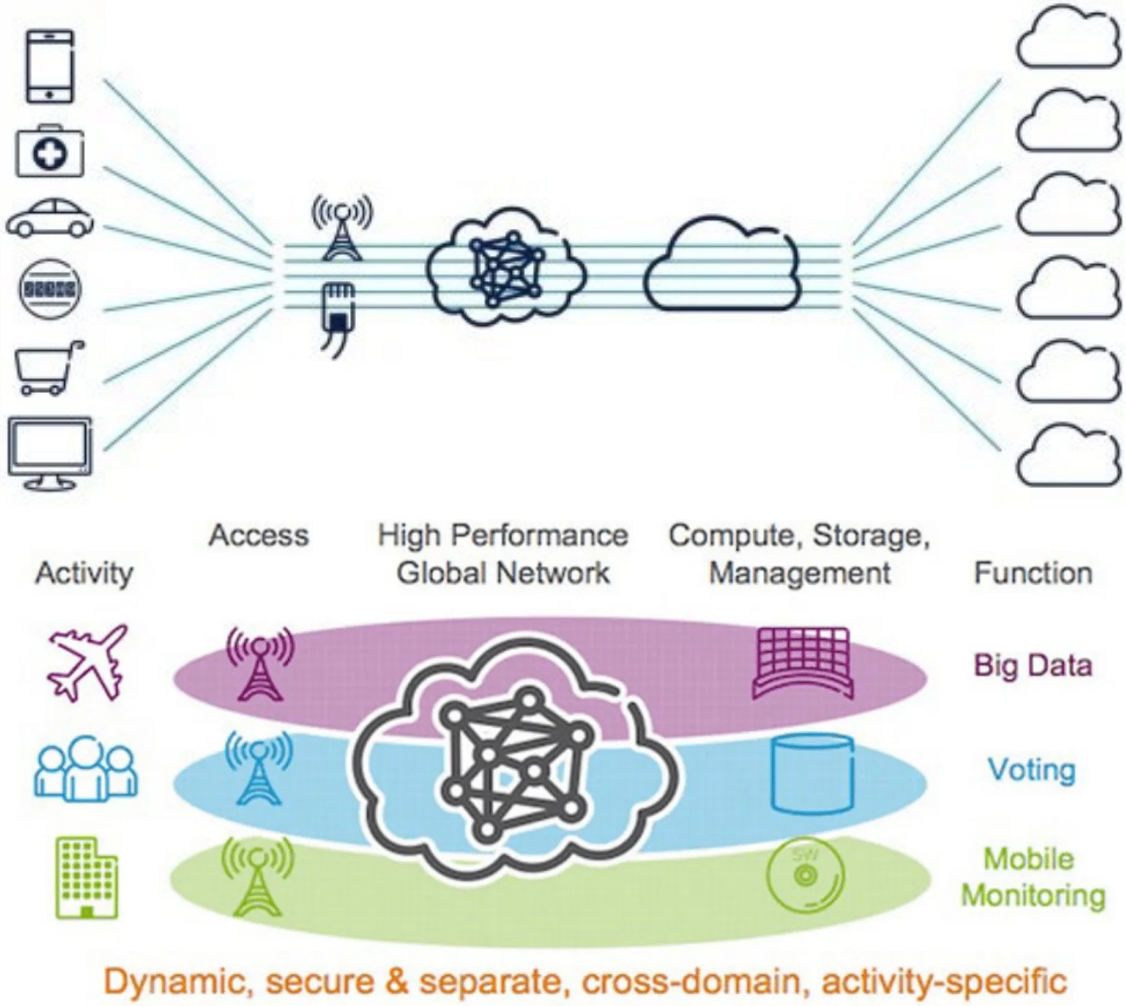


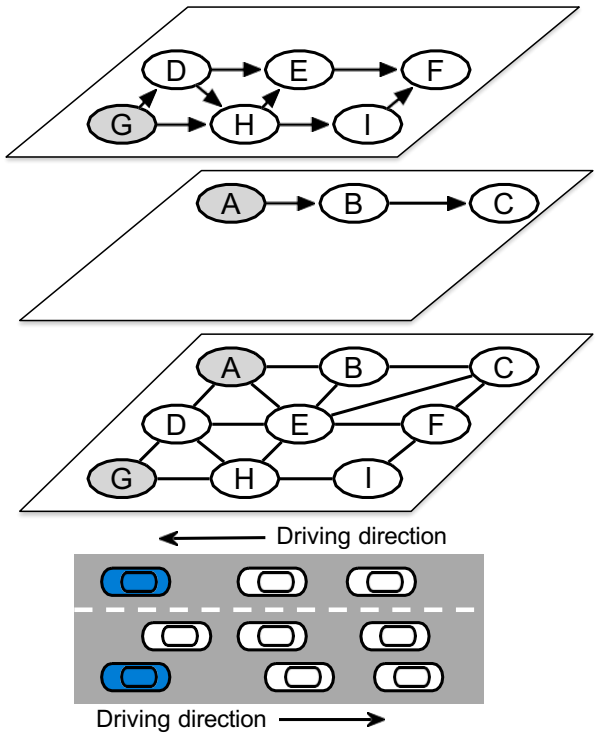


SDN Enables Network Slicing

Network slicing is a concept that network architecture is able to be partitioned on demand to accommodate various use cases.

Allow operators to design, deploy, and customize these network slices atop a common network infrastructure





(d) Network Slice 2: Emergency vehicle passing request

(c) Network Slice 1: Sudden brake warning

(b) Vehicular Network Topology

(a) Physical Location

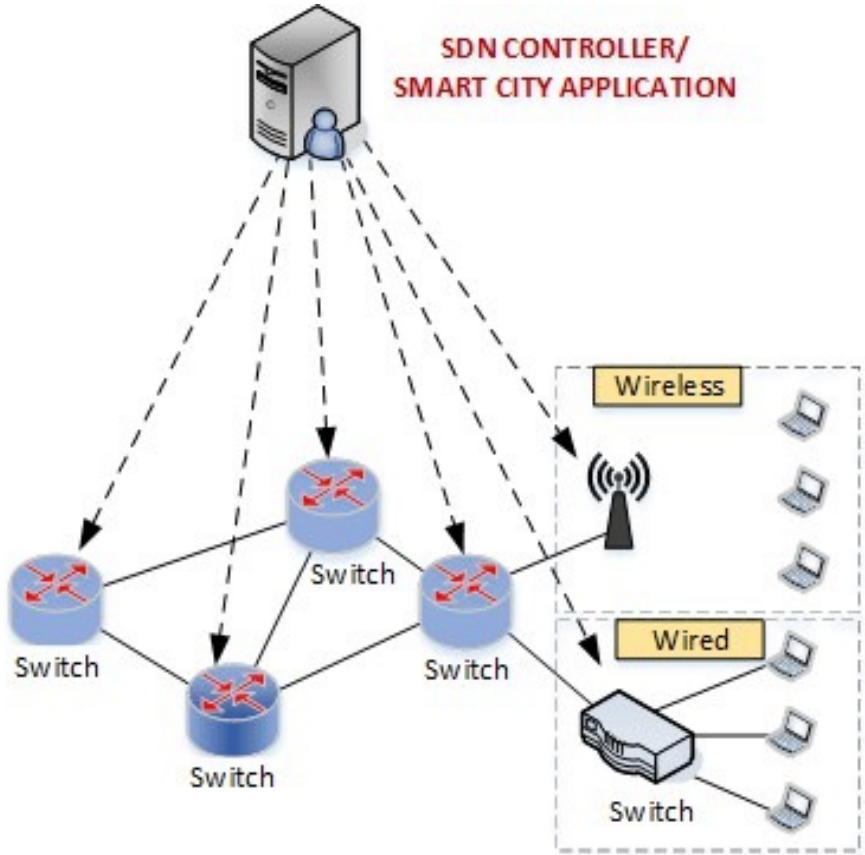
SDN Supports Network Virtualization

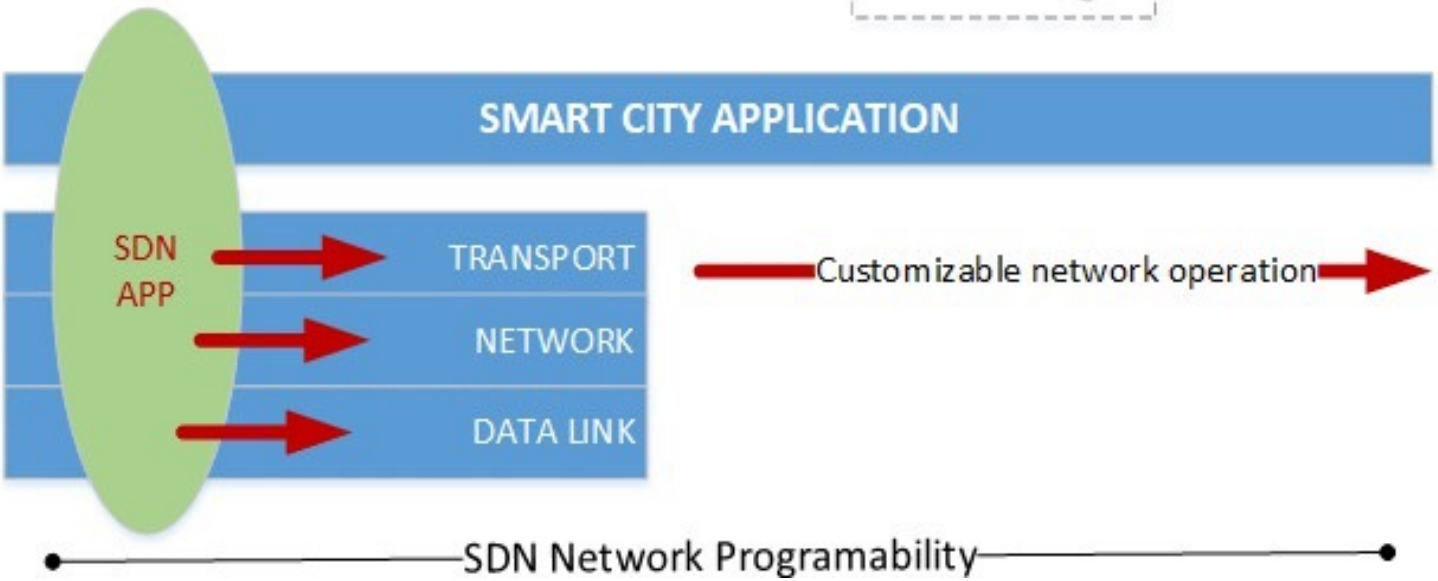
SDN supports network virtualization, and programmability

The ability to improve network flexibility, including the ability to dynamically change configurations, traffic flows and provide application-level QoS.

How?

Controller enables new / different control of forwarding, traffic flow, and protocols in data center fabrics and other cloud networks.





Desktop Virtualization

Abstract the hardware, OS, applications, data, and user profile of a real desktop from a client machine to a remote server.

When user accesses a client desktop, all applications and corresponding data are accessed from the cloud server.

Enables IT organizations to respond faster to changing workplace needs and emerging opportunities

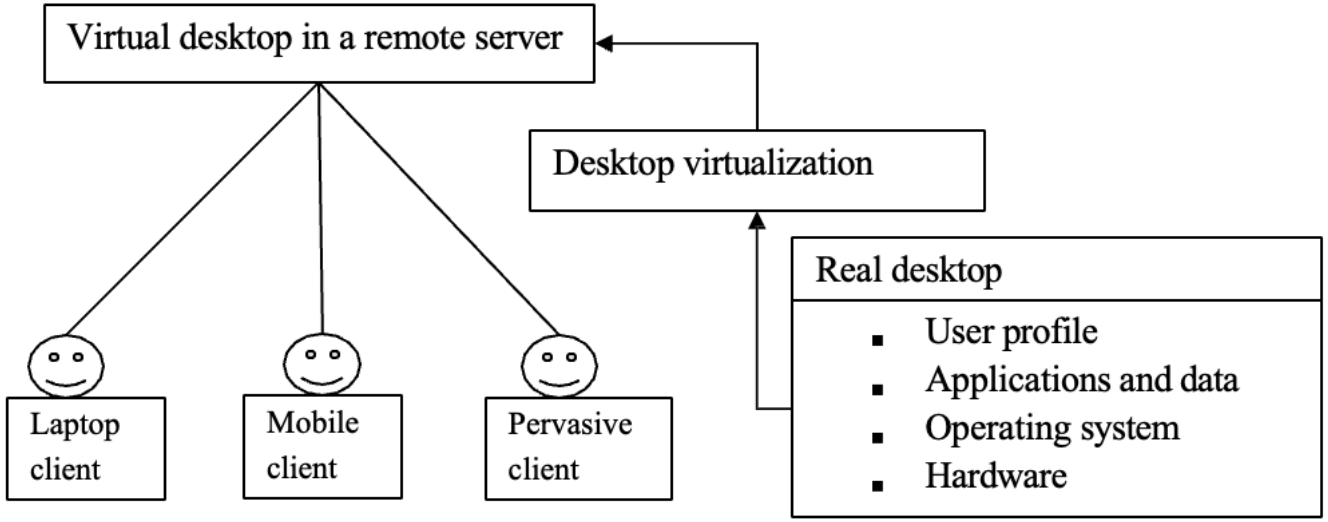
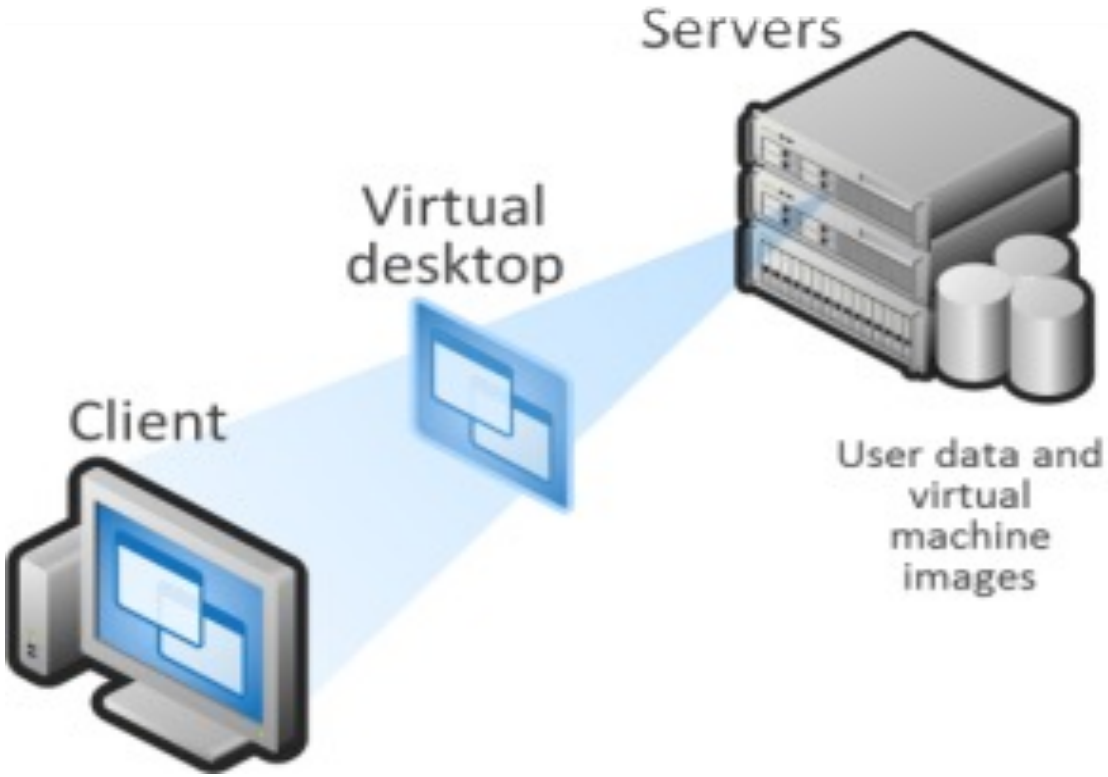


Fig. 3.20 Desktop virtualization



Data Virtualization

Aggregating data from heterogeneous data sources at different locations into logical and single items of data that can be accessed easily by applications

Provide federated data

Provide a single data management model

Facilitate data access, data integration, and data management

Retrieve and manipulate data from one or more data sources without knowing the technical details of data storage such as API, access language, storage structure, or its format or location

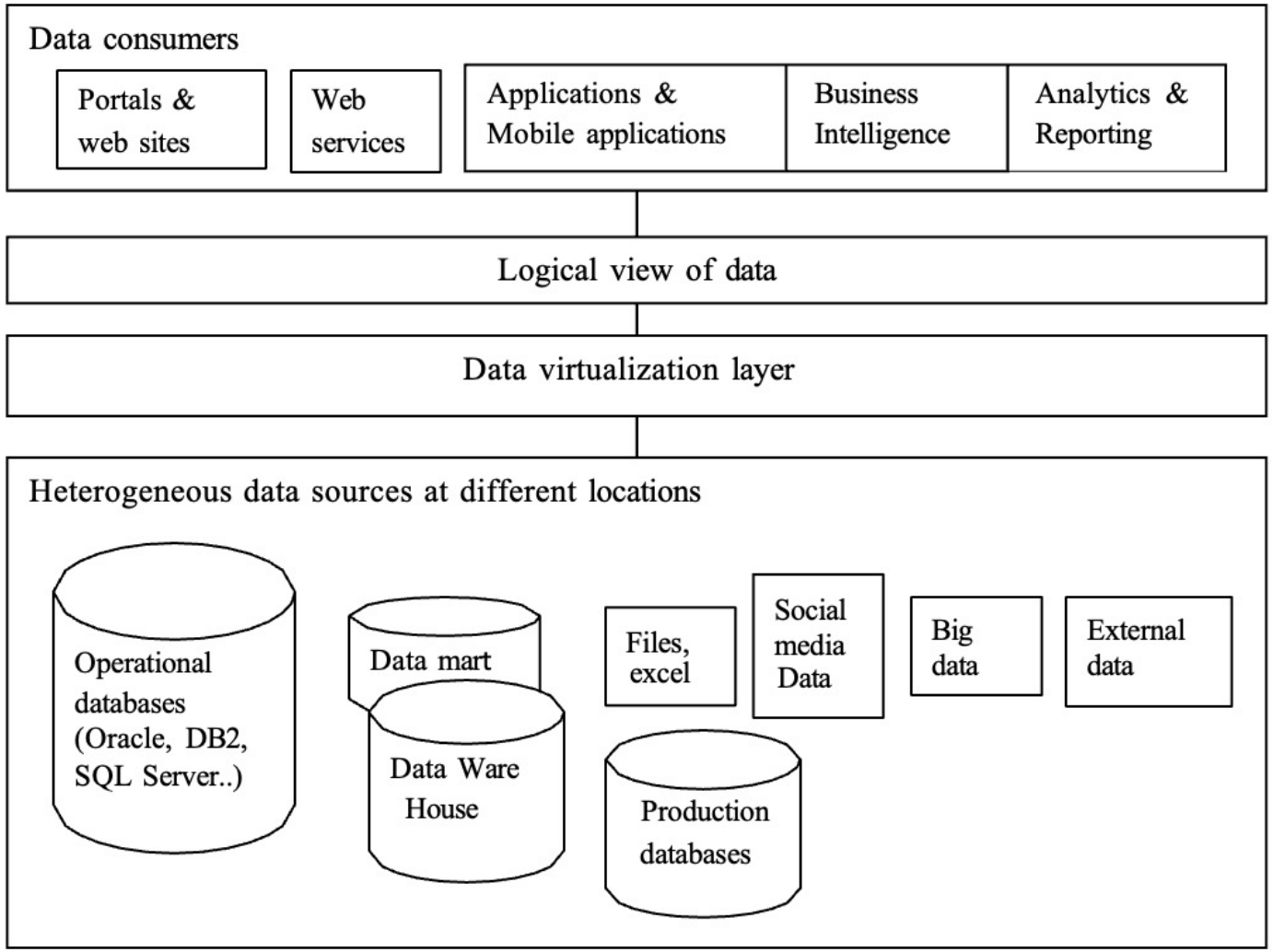


Fig. 3.21 Data virtualization

Application Virtualization

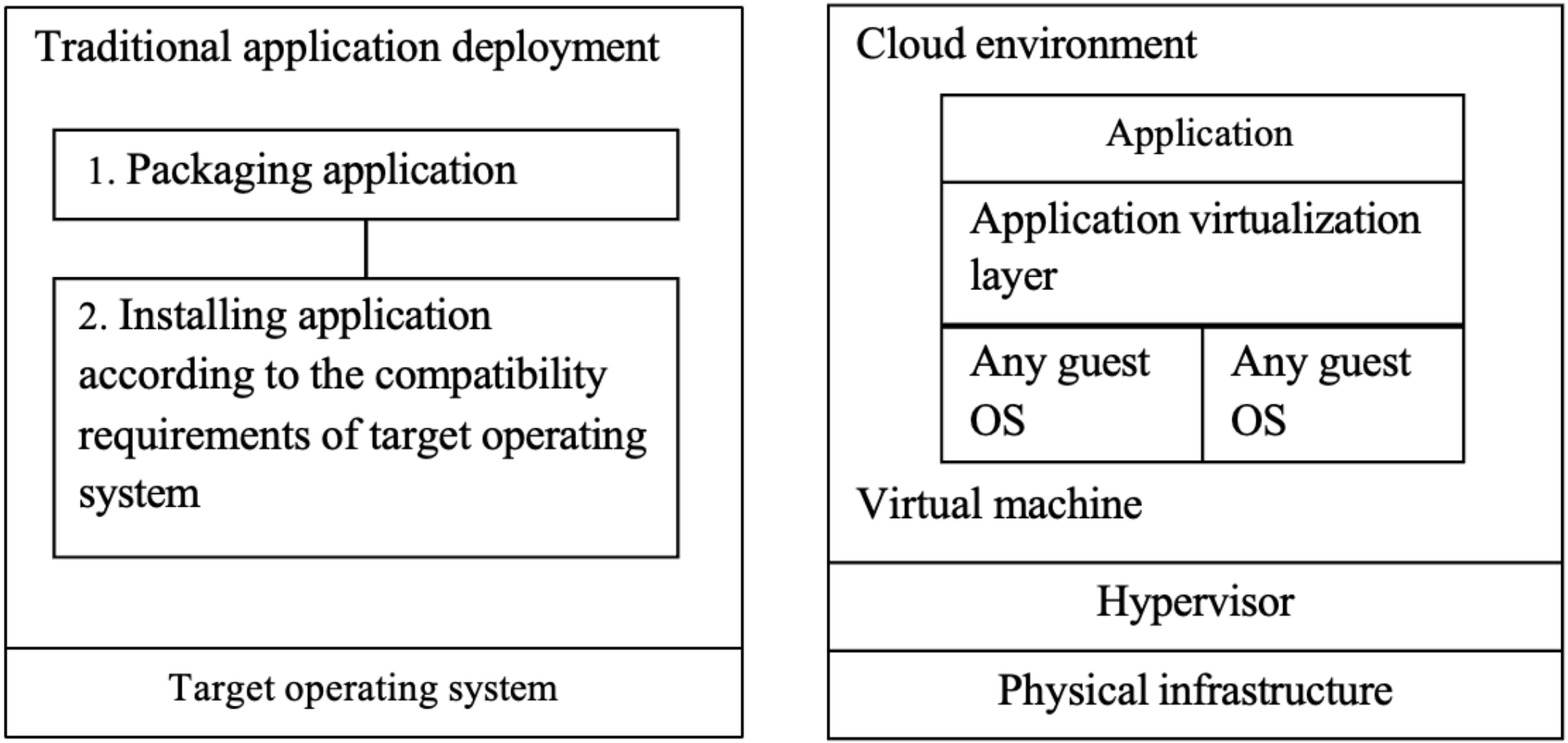


Fig. 3.22 Application deployment in traditional and cloud environment